

TOPIC 6: AIR AND WATER POLLUTION

1. Sources of Air Pollution

Vehicle exhaust, factory smoke, burning of refuse and bush, dust, and gases from generators and cooking fuels. Pollutants include carbon monoxide, sulphur dioxide, nitrogen oxides and particulate matter.

2. Sources of Water Pollution

Sewage, industrial waste, agricultural runoff (fertilisers and pesticides), oil spills, and dumping of refuse into rivers and drains.

3. Effects on Living Organisms

Air pollution: breathing problems, asthma, eye irritation, acid rain. Water pollution: waterborne diseases (cholera, typhoid), death of fish and aquatic plants, unsafe drinking water.

4. Prevention and Purification

Air: cleaner fuels, vehicle maintenance, reduce open burning, plant trees. Water: treat sewage, avoid dumping waste in water bodies, protect catchment areas. Household purification: boiling, filtration, chlorination, settling.

5. Simple Experiment Idea

Filtration of muddy water: pour muddy water through filter paper or a cloth and sand/charcoal layers; observe clearer water. Discuss that further treatment may still be needed for drinking.

6. Summary

Air and water pollution come from human activities and harm health and ecosystems. Prevention and purification are essential.

End of Topic 6